AI ACADEMY · LEVEL 7 · GRADE 12 · WEEK 1 · WORKBOOK

A question that evidence can answer

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

One complete question, outcome and condition record.

Keep comparison, cases, scoring, fixed conditions, available evidence and remaining gaps together.

Keep one habit: Recall → W1–W3/W5 → independent timing/correctness case → research-question project → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Engineering demonstration and defense. Give an example and a mistake that the idea helps you avoid.

Engineering demonstration and defense

> **My remembered idea, example and correction**

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- An answerable research question identifies outcome, comparison and conditions. Replace "prove AI is useful" with a bounded retrieval question: on fixed queries Q1–Q5 and corpus c1 R1–R10, does B return more supplied-relevant records than baseline A in the top three on average? Fix relevance sets, unique eligible top-three lists and case IDs before comparison.

  Added revealed practice snapshot has A counts 1/1/1/1/1 and B 2/1/2/0/2: means 1 and 1.4 relevant records, with Q4 worse. These rankings/labels are constructed, not live retrieval or human judgments; practice is not unseen reserved evaluation. Separate picture compares diagram/no-diagram A/B on the same six fixed plant tasks and scoring; no results are supplied.

  "Wonderful" has no measurement here; correct-answer count has one. Easier B-only tasks confound difficulty with version. A bounded result does not establish universal usefulness, causal mechanism, deployment impact or effects on children/general learning. Keep planned study, constructed example, actual scoring run and uncollected outcomes distinct.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

MAIN LESSON · SECTION 6

W2 · Trace the worked case

> Keep the supplied inputs, comparison rule and evidence together.

Complete task and evidence

- An answerable research question identifies outcome, comparison and conditions. Replace “prove AI is useful” with a bounded retrieval question: on fixed queries Q1–Q5 and corpus c1 R1–R10, does B return more supplied-relevant records than baseline A in the top three on average? Fix relevance sets, unique eligible top-three lists and case IDs before comparison.

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- Proposal says prove AI is useful; available records are retrieval results for five room queries. Show the intermediate steps and give the conclusion with its **scope**.

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- An answerable research question identifies outcome, comparison and conditions. Replace “prove AI is useful” with a bounded retrieval question: on fixed queries Q1–Q5 and corpus c1 R1–R10, does B return more supplied-relevant records than baseline A in the top three on average? Fix relevance sets, unique eligible top-three lists and case IDs before comparison.

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- A report omits this qualification: A measurable narrow question is not **evidence** of its answer. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- An answerable research question identifies outcome, comparison and conditions. Replace "prove AI is useful" with a bounded retrieval question: on fixed queries Q1–Q5 and corpus c1 R1–R10, does B return more supplied-relevant records than baseline A in the top three on average? Fix relevance sets, unique eligible top-three lists and case IDs before comparison.

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- For the worked scenario, propose one additional case that could reveal a failure, describe its expected **outcome** before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- You have response times and correctness for two lookup procedures on ten fictional requests. Write a bounded question and one missing detail.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

My independent explanation, trace and limits

MAIN LESSON · SECTION 10

Project · Keep the research evidence

> Preserve the question, protocol, observations, analysis and limits.

Complete task and evidence

- Add a labelled a question that **evidence** can answer **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of a question that **evidence** can answer.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for A question that **evidence** can answer. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for A question that **evidence** can answer. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- You have response times and correctness for two lookup procedures on ten fictional requests. Write a bounded question and one missing detail.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

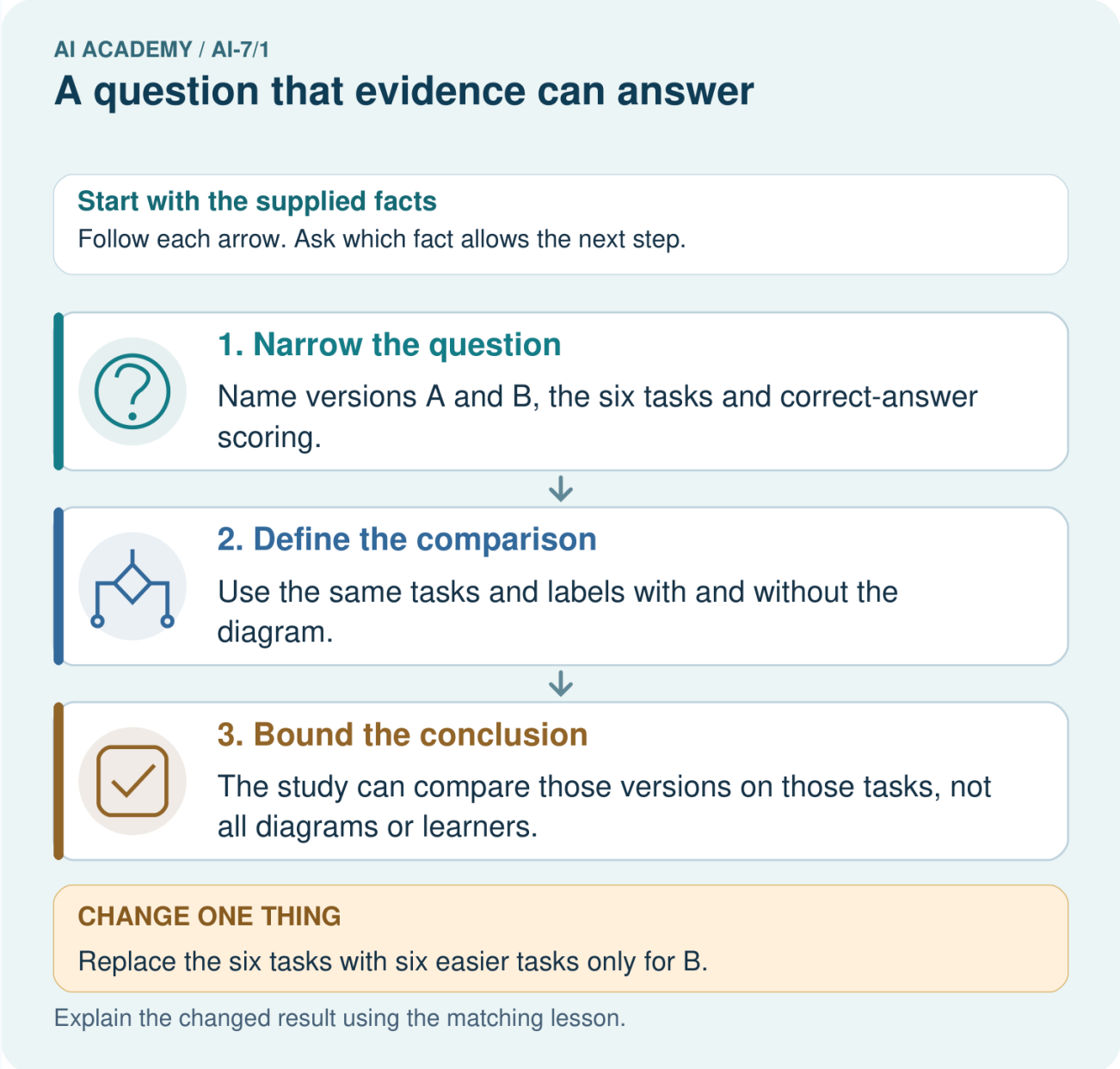
- A question with a measurable answer. A team wants to know whether giving a diagram changes accuracy on six fixed plant-identification tasks.

- Candidate questions: "Are diagrams wonderful?" and "Does version B answer more of these six tasks correctly than version A?"

- Which proposed diagram question has a measurable **outcome**? Name its versions, six-case denominator and fixed scoring, then explain why giving only B easier tasks spoils the comparison.

- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

  Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project milestone**

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Reproducible AI Research Study
- Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.
- For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- A question with a measurable answer. A team wants to know whether giving a diagram changes accuracy on six fixed plant-identification tasks.
- Which labelled step matches this detail? The study can compare those versions on those tasks, not all diagrams or learners.
- 1. Define the comparison
- 2. Bound the conclusion

- 3. Narrow the question

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Bound the conclusion The study can compare those versions on those tasks, not all diagrams or learners.

An answerable **research question** connects a change to an observable **outcome** and a specified population. Opinion words need an operational definition before measurement.

Changed-case question and optional hint

- Changed-case practice: Replace the six tasks with six easier tasks only for B. Which conclusion is supported by the supplied facts?

- 1. This one example guarantees correct results for every future **input**.

- 2. The comparison no longer isolates the diagram condition.

- 3. We can decide the result without examining the changed **input** or the procedure.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The comparison no longer isolates the diagram condition. Task difficulty now changes alongside the proposed intervention.

Repair and retain the record

Review Engineering demonstration and defense (ai-6/36). Goal: Explain the **mechanism** and terminology of engineering demonstration and defense. Short practice: Locate V7, room E and read-only **scope** in the demonstration.

Which **evidence** supports the answer, and why must a request to email everyone still be refused?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

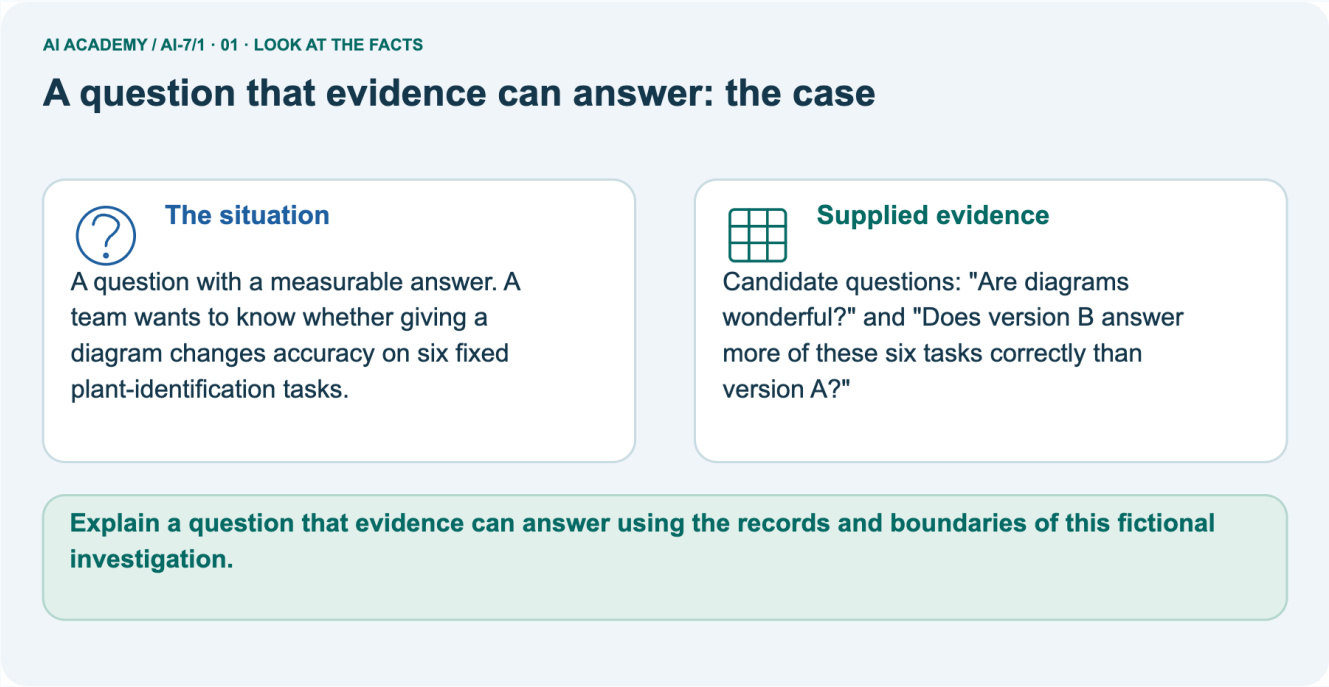
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. A question that evidence can answer: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

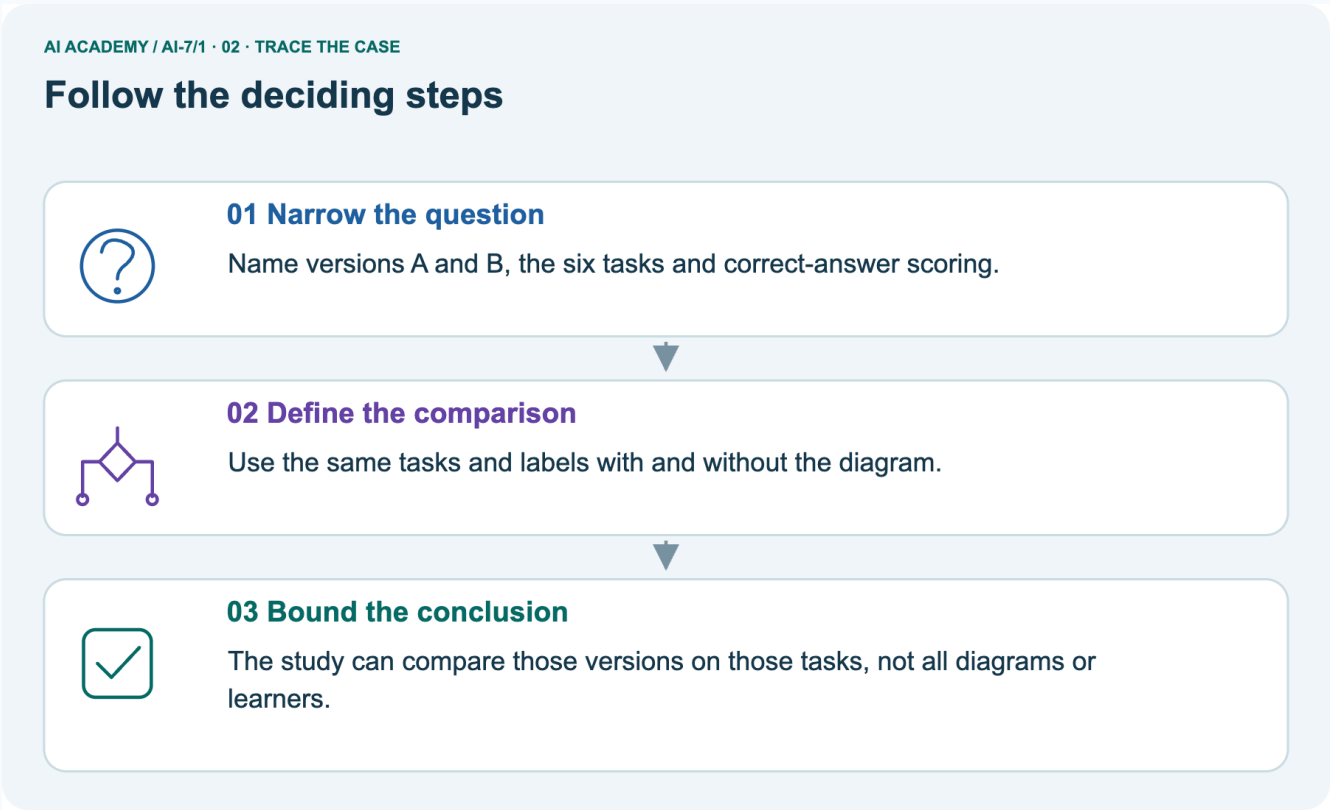
A question with a measurable answer. A team wants to know whether giving a diagram changes accuracy on six fixed plant-identification tasks.

Supplied facts: Candidate questions: "Are diagrams wonderful?" and "Does version B answer more of these six tasks correctly than version A?"

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



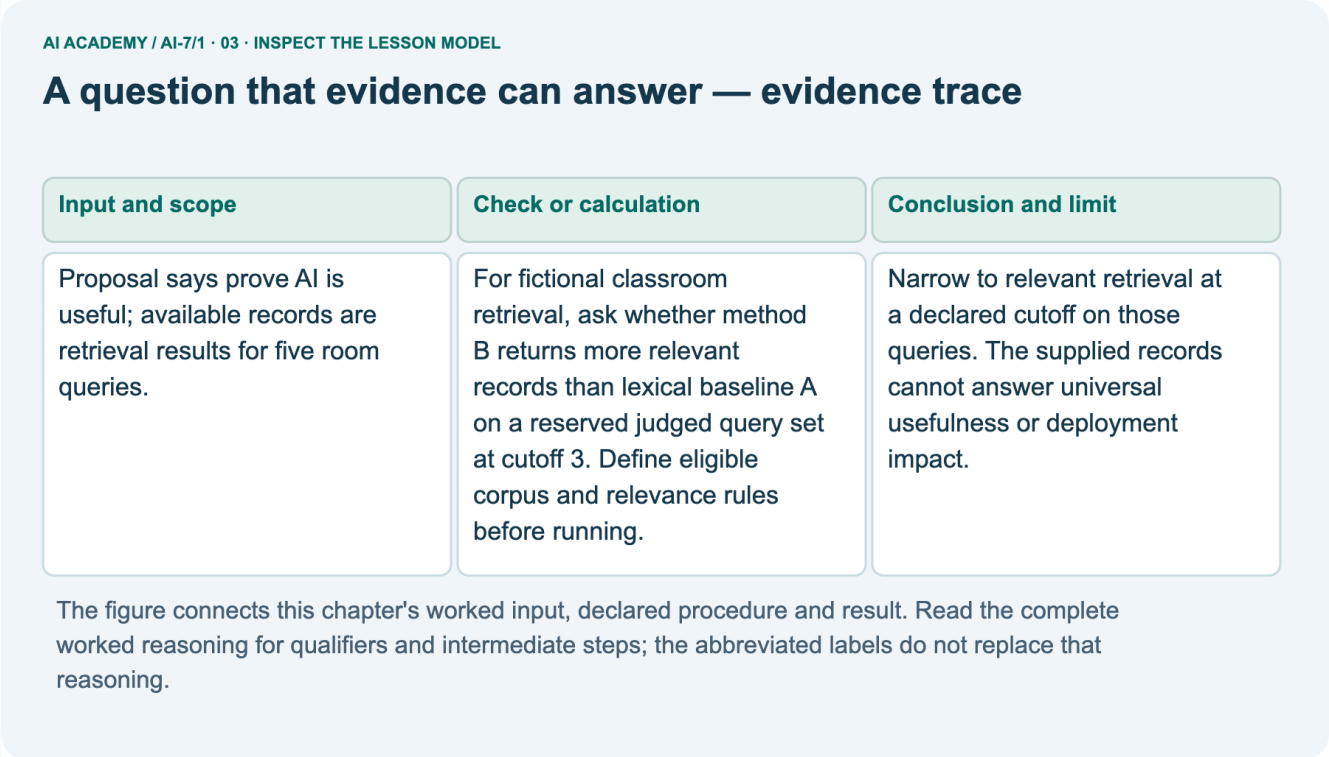
Follow the labelled picture with the matching explanation. Open the vector diagram

Narrow the question: Name versions A and B, the six tasks and correct-answer scoring. → Define the comparison: Use the same tasks and labels with and without the diagram. → Bound the conclusion: The study can compare those versions on those tasks, not all diagrams or learners.

Point to each numbered stage and explain which recorded **input** it uses.

> **My picture explanation, deciding evidence and one limit**

03. A question that evidence can answer — evidence trace



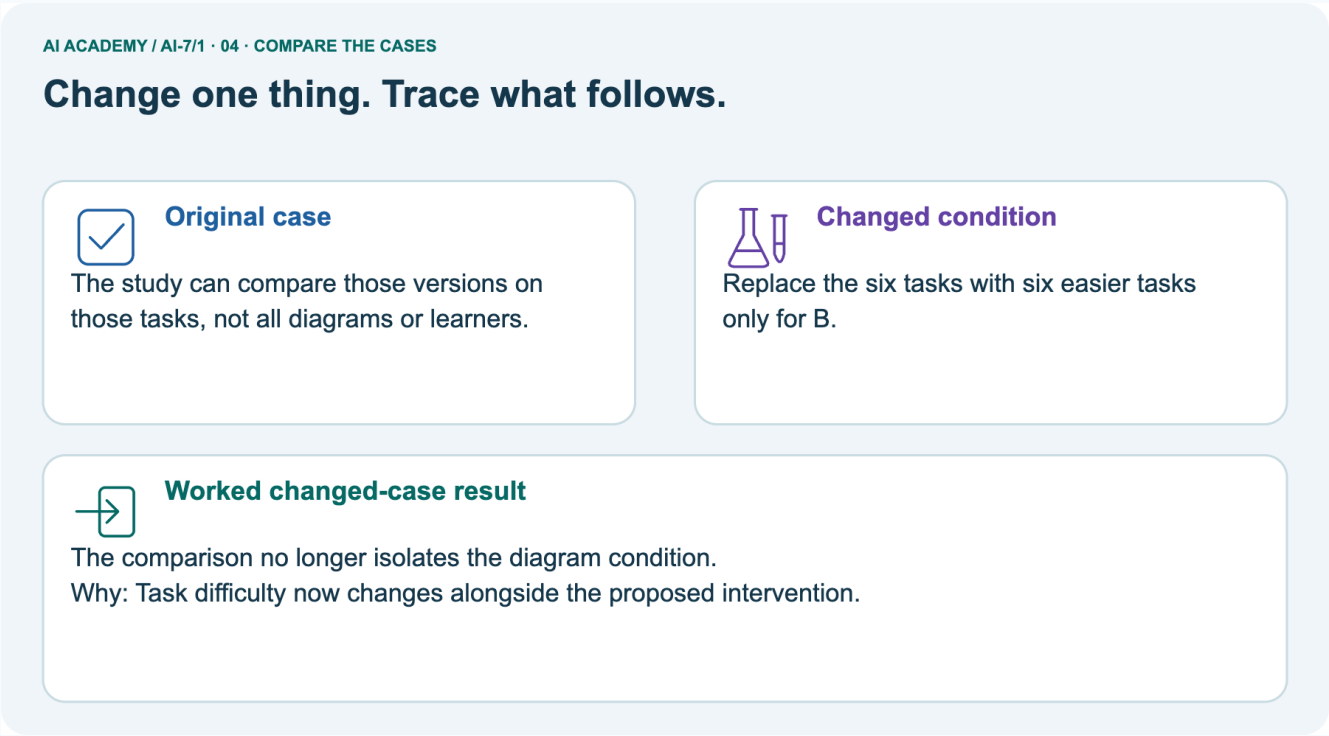
Follow the labelled picture with the matching explanation. Open the vector diagram

A question that **evidence** can answer — **evidence** trace. Proposal says prove AI is useful; available records are retrieval results for five room queries. | For fictional classroom retrieval, ask whether method B returns more relevant records than lexical baseline A on a reserved judged query set at cutoff 3. Define eligible corpus and relevance rules before running. | Narrow to relevant retrieval at a declared cutoff on those queries. The supplied records cannot answer universal usefulness or deployment impact.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

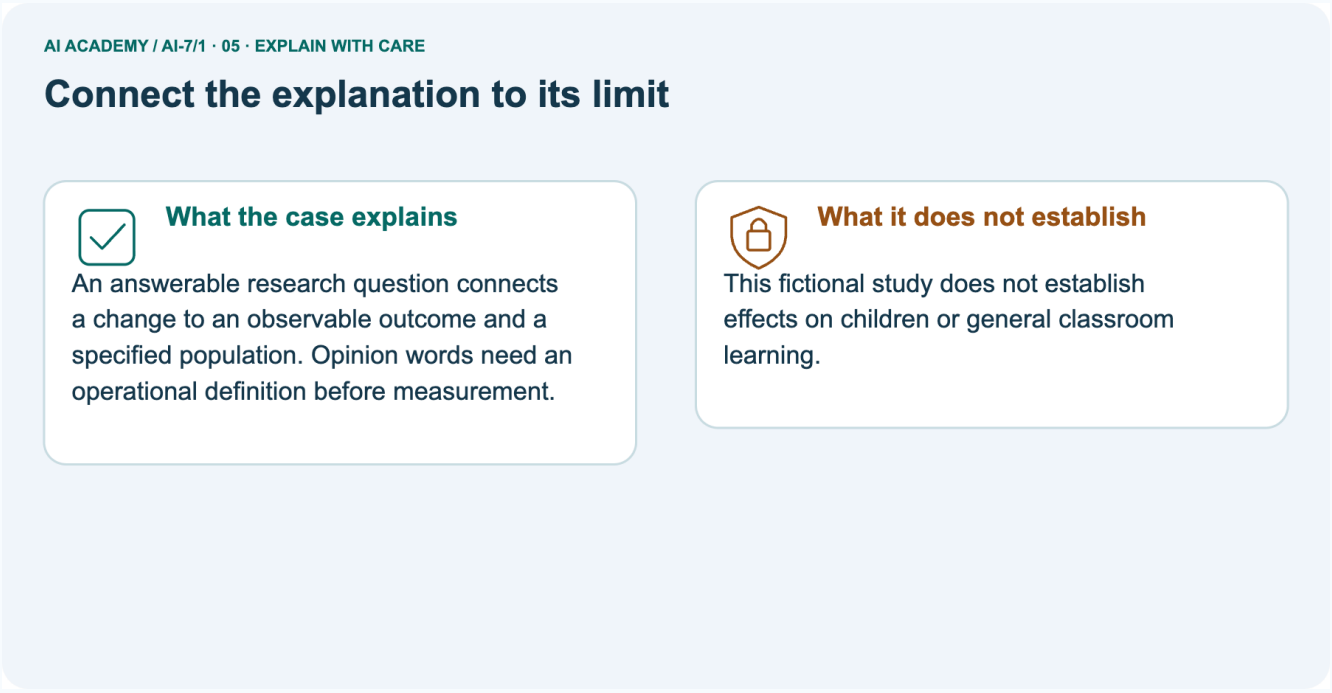
Original case: The study can compare those versions on those tasks, not all diagrams or learners. Changed condition: Replace the six tasks with six easier tasks only for B.

Expected result: The comparison no longer isolates the diagram condition. Reason: Task difficulty now changes alongside the proposed intervention.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

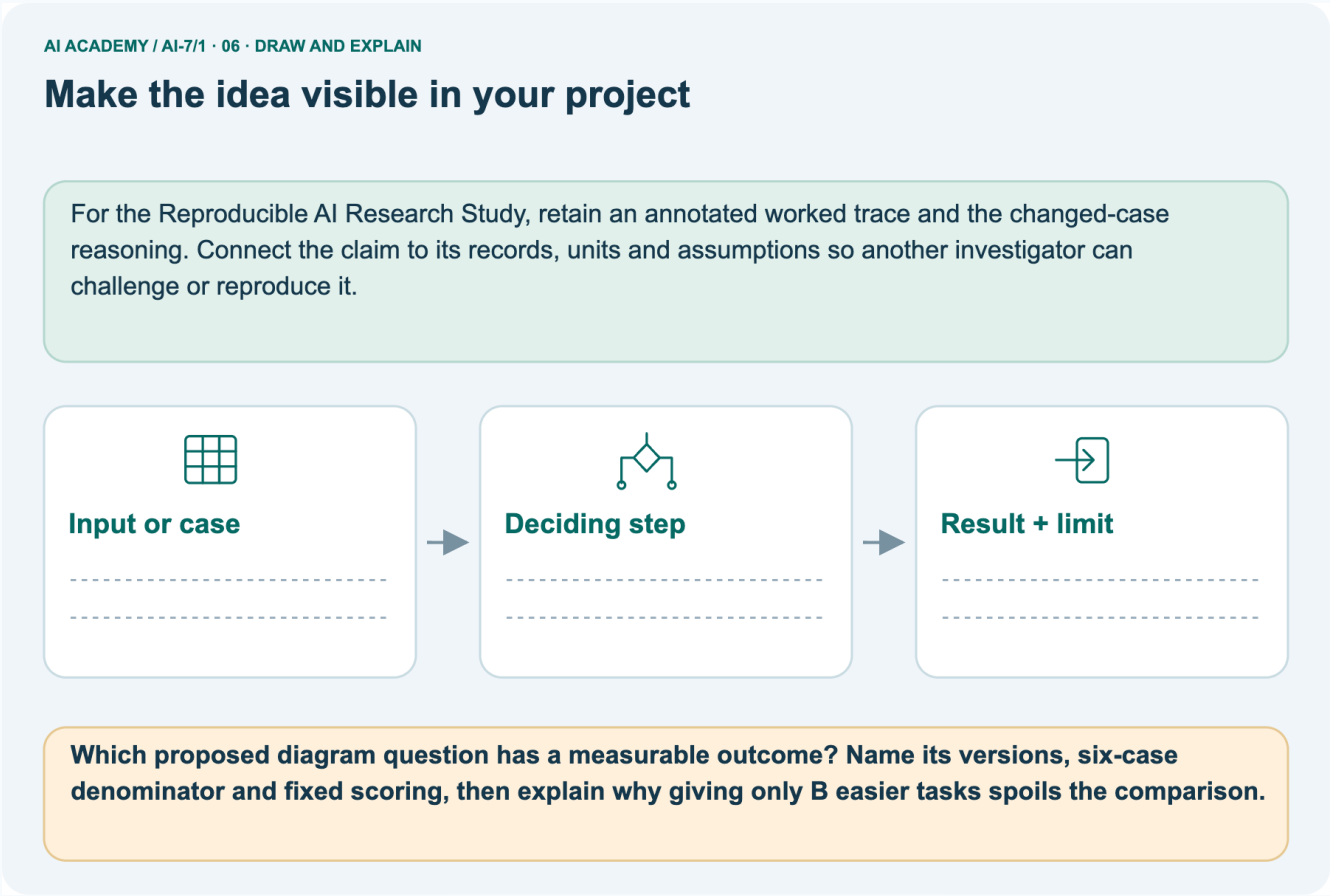
An answerable **research question** connects a change to an observable **outcome** and a specified population. Opinion words need an operational definition before measurement.

Limit: This fictional study does not establish effects on children or general classroom learning.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Which proposed diagram question has a measurable **outcome**?

Name its versions, six-case denominator and fixed scoring, then explain why giving only B easier tasks spoils the comparison.

Draw your project trace and justify the deciding step using **evidence**.

> **My picture explanation, deciding evidence and one limit**

▼ **Earlier wording, alternatives and picture transcripts**

01. A question that evidence can answer: the case

02. Follow the deciding steps

03. A question that evidence can answer — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ **Optional interactive practice and completed work**

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.